

Sequence Check-In [40 marks]

1. [Maximum mark: 6] 25N.1.SL.TZ1.2

The 1st and 5th terms of an arithmetic sequence are 36 and 12 respectively.

- (a) Find the 13th term of this arithmetic sequence. [4]

The sum of the first n terms of this arithmetic sequence is zero.

- (b) Find the value of n . [2]

2. [Maximum mark: 6] 24N.1.AHL.TZ0.5

For a particular arithmetic sequence, $u_{10} = 16$ and $S_{25} = 100$.

Find the value of k such that $u_k = 0$. [6]

3. [Maximum mark: 16] 25M.1.SL.TZ3.8

Consider the sequence $\{u_n\}$, with n th term given by u_n . The first three terms are

$$u_1 = k - 5, u_2 = 3 - 2k \text{ and } u_3 = 5k + 3, \text{ where } k \in \mathbb{R}.$$

- (a) Consider the case when $\{u_n\}$ is arithmetic.

- (a.i) Find the value of k . [3]

- (a.ii) Hence, or otherwise, find u_3 . [2]

- (b) Consider the case where $k = 12$.

- (b.i) Show that the first three terms of $\{u_n\}$ form a geometric sequence. [3]

- (b.ii) Given that $\{u_n\}$ is geometric, state a reason why the sum of an infinite number of terms of this sequence does not exist. [1]
- (c) The sequence, $\{u_n\}$, is geometric for a second value of k .
- (c.i) Show that $k^2 - 10k - 24 = 0$. [2]
- (c.ii) Find the first three terms of $\{u_n\}$ for this second value of k . [4]
- (c.iii) Hence, write down the value of S_{2m} , the sum of the first $2m$ terms, for this second value of k . [1]

4. [Maximum mark: 7] 25M.2.AHL.TZ2.7
A geometric sequence has first term 80 and fourth term 0.74088.

- (a) Find the second term. [3]

The first two terms of this geometric sequence are also the first term and eleventh term respectively, of an arithmetic sequence.

Let S_n denote the sum of the first n terms of the arithmetic sequence.

- (b) Find the greatest value of S_n , giving your answer to two decimal places. [4]

5. [Maximum mark: 5] 25N.2.AHL.TZ1.6
Consider the infinite series

$$\sum_{k=0}^{\infty} \left(\frac{53}{1000}\right) \left(\frac{1}{100}\right)^k.$$

- (a) Find the exact value of S_{∞} . [3]

Let $0.\overline{253}$ represent the repeating decimal $0.2535353\ldots$

- (b) Use your answer from part (a) to express $0.\overline{253}$ as a fraction such that the numerator and denominator have no common factors.

[2]